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Title

Extending CWIIF: Identification of Influential Nodes in Weighted and Directed Multilayer Networks Using Coupled Intra-Layer and Inter-Layer Influence Factors

Problem Statement

Identifying influential nodes is a fundamental problem in network science, with direct applications in controlling epidemic outbreaks, optimizing viral marketing, and securing critical infrastructure. Real-world systems, however, are rarely single-layered; entities such as people, genes, or transit stations interact simultaneously through multiple, structurally distinct layers (for example, social, professional, and digital ties for the same individual). Existing node-ranking methods handle this complexity poorly: many either flatten all layers into one aggregated network, discarding vital structural information, or treat every layer as equally important, ignoring the fact that some layers are intrinsically more efficient at spreading influence than others.

The recently proposed CWIIF (Coupling Weighted Intra-layer and Inter-layer Influence Factors) algorithm addresses this gap by combining layer-specific weighting (based on active nodes, active paths, and inter-layer edge overlap) with node-level centrality (K-shell decomposition fused with betweenness centrality) to compute a single, reliable influence score per node. While CWIIF outperforms nine classical and state-of-the-art benchmarks on real-world datasets, it is restricted to unweighted, undirected multilayer networks and carries a cubic time complexity that limits its scalability to very large networks. This project addresses this technology gap by first implementing and empirically validating the CWIIF algorithm on benchmark multilayer datasets, and then extending its formulation to weighted and/or directed multilayer networks, while exploring optimizations to improve its computational efficiency for larger, real-world graphs.

Abstract

The rise of interconnected digital and physical systems — social platforms, transportation networks, financial systems, and biological pathways — has shifted network science research from single-layer graphs toward multilayer networks, where the same set of nodes is connected through several structurally distinct layers. Accurately ranking node influence in such systems is significantly harder than in single-layer networks: a node may be a critical hub in one layer while being entirely peripheral in another, and different layers can differ sharply in how efficiently they propagate information. Traditional approaches either aggregate all layers into a single network, losing dimensional information, or assign equal importance to every layer, which contradicts the reality that layer topology strongly affects spreading speed.

This project studies the CWIIF algorithm, a recent method that computes a layer-weight coefficient for every layer using three normalized factors — active nodes, active path length, and Jaccard-based inter-layer communication intersection — and separately computes a node's intra-layer influence by fusing normalized K-shell values with normalized betweenness centrality. These two components are coupled through a weighted summation to produce a single Total Influence (TI) score per node, enabling a global ranking across the entire multilayer network. The reported results show CWIIF outperforming nine classical and state-of-the-art baselines (sumPR, sumBC, sumDC, aggKS, M-H, SVT, mlPCI, MSV, and WGC) on nine real-world multilayer datasets and one synthetic Barabási–Albert network, evaluated using Spearman and Kendall rank correlation against Susceptible-Infected (SI) spreading simulations, the imprecision function, the monotonicity index, and network-connectivity-based robustness testing.

In this project, we will first reproduce these experiments to build a working baseline and confirm CWIIF's reported advantages. We will then address two limitations acknowledged by the original authors: (i) the algorithm's restriction to unweighted, undirected networks, and (ii) its $O(N^3)$ time complexity, which is dominated by all-pairs shortest-path computation for the active-path and betweenness-centrality factors. We propose to generalize the layer-weighting and node-influence formulations to accommodate edge weights and directionality, and to evaluate approximate or sampling-based alternatives to exact shortest-path and betweenness computation to improve scalability, while preserving ranking accuracy on the same evaluation metrics used in the original study.

Proposed Approach

Our methodology begins with acquiring and preprocessing the benchmark real-world multilayer datasets used in the CWIIF study (social, transport, academic co-authorship, and biological networks), sourced from public multilayer network repositories, along with a synthetic Barabási–Albert multilayer network for scalability testing. We will implement the three-phase CWIIF pipeline in Python: (1) computing layer-weighted coefficients from normalized active-node, active-path, and inter-layer Jaccard-intersection factors; (2) computing each node's intra-layer influence vector by summing normalized K-shell and betweenness centrality values per layer; and (3) mapping these vectors into a single Total Influence score per node using the layer weights.

We will validate our implementation against the original paper's reported results using Spearman/Kendall correlation with Susceptible-Infected (SI) spreading simulations, the imprecision function, the monotonicity index, and targeted-node-removal connectivity tests. Once the baseline is verified, we will extend the mathematical formulation to support weighted and directed edges, adapt the K-shell and betweenness computations accordingly, and evaluate the extended algorithm on weighted/directed variants of the benchmark datasets. Finally, we will explore complexity-reduction strategies (e.g., approximate betweenness centrality, layer-sampling for path computation) and benchmark the trade-off between computational cost and ranking accuracy.

Objectives

- To study the CWIIF algorithm in depth and implement a working baseline on standard benchmark multilayer network datasets to reproduce and validate the originally reported results.

- To evaluate the implemented algorithm using Spearman and Kendall rank correlation (against SI-model spreading simulations), the imprecision function, the monotonicity index, and network-connectivity-based robustness testing, benchmarked against existing methods.
- To extend the CWIIF framework's layer-weighting and intra-layer influence formulations to support weighted and/or directed multilayer networks.
- To investigate optimization techniques (e.g., approximate shortest-path/betweenness computation) that reduce the algorithm's time complexity for large-scale networks, and to evaluate the resulting accuracy–efficiency trade-off.

Keywords

Multilayer Networks, Influential Node Identification, K-shell Decomposition, Betweenness Centrality, Complex Networks, Network Science, Susceptible-Infected Model, Layer Weighting.

Technologies and Tools to be Used

Dataset: Publicly available multilayer network datasets (Vickers-Chan-7thGraders, Kapferer-Tailor-Shop, Lazega-Law-Firm, CS-Aarhus, London Transport, PierreAuger Co-authorship, Drosophila, Sacchpomb, and Homo Genetic multiplex networks; source: manliodedomenico.com/data.php), along with a synthetic Barabási–Albert multilayer network.

Language: Python (primary).

Libraries: NetworkX, NumPy, SciPy, Matplotlib/Seaborn for analysis and visualization.

Software Requirements: Windows 11 (64-bit), Jupyter Notebook, Google Colab.

Hardware Requirements: Intel i5-class processor (or equivalent) with integrated/discrete GPU optional.

Other Software: MS Excel / Google Sheets (for logging and result tables).

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